Practical No 4

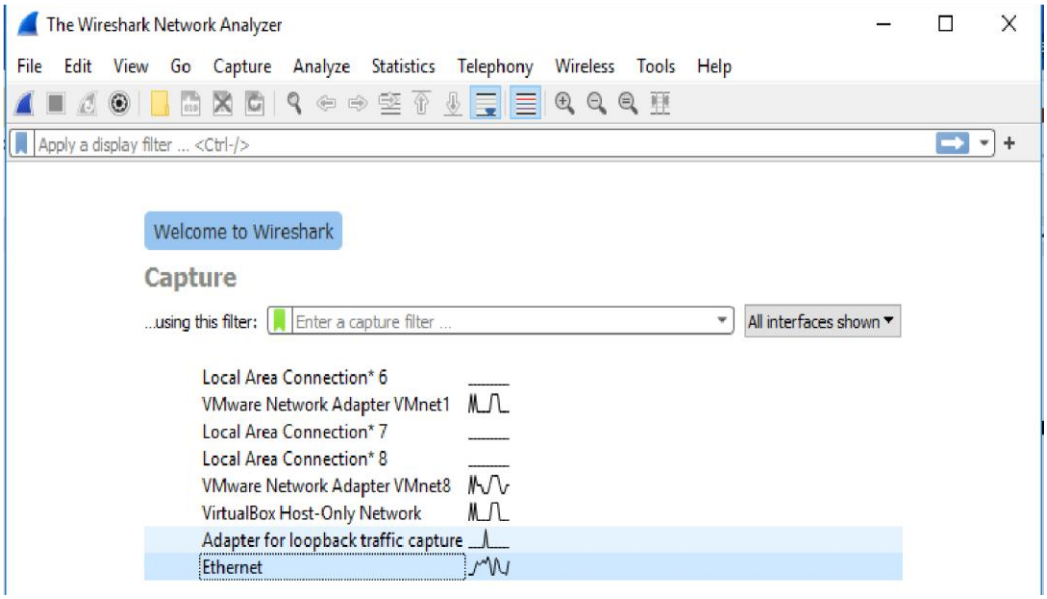
Aim: Capturing and analysing network packets using Wireshark (Fundamentals):

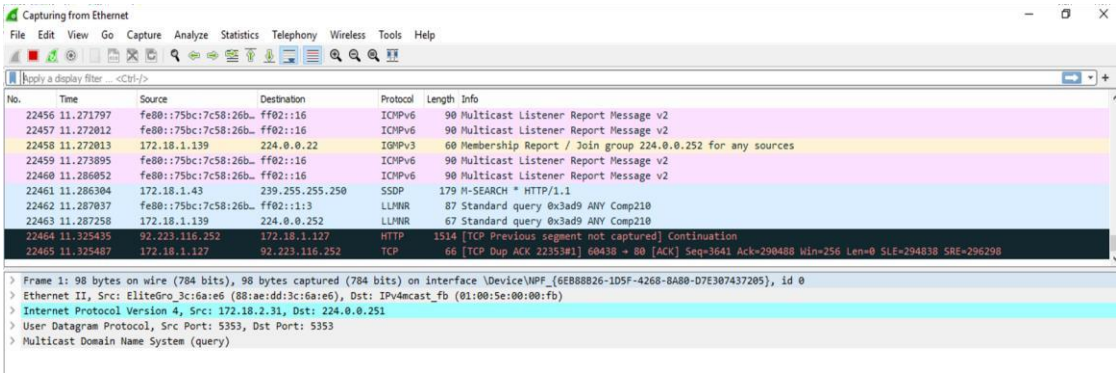
- **Identification of the live network**
- **Capture Packets**
- **Analyse the captured packets.**

Steps:

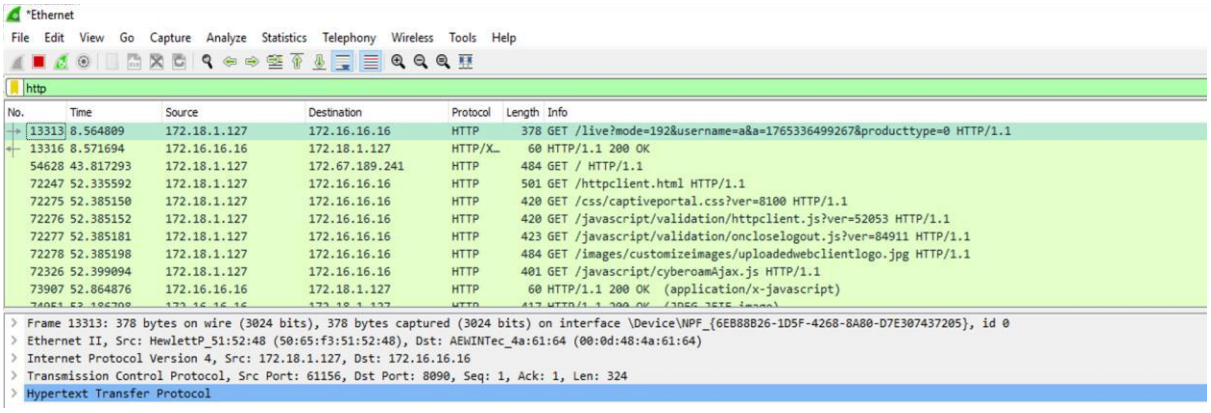
Capturing Packets

1. Open Wireshark and click on Ethernet.

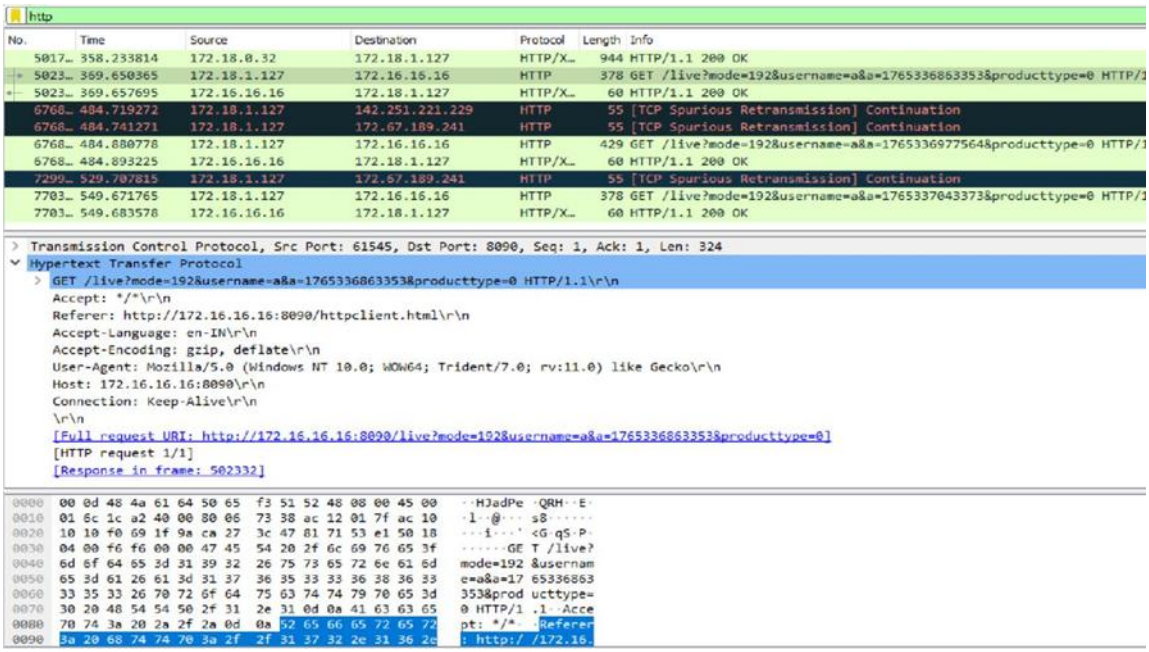




2. Go to browser and open any unsecured website like www.gmail.com.
3. Perform some activity on the website.
4. Now, go back to Wireshark and enter http in the search bar.

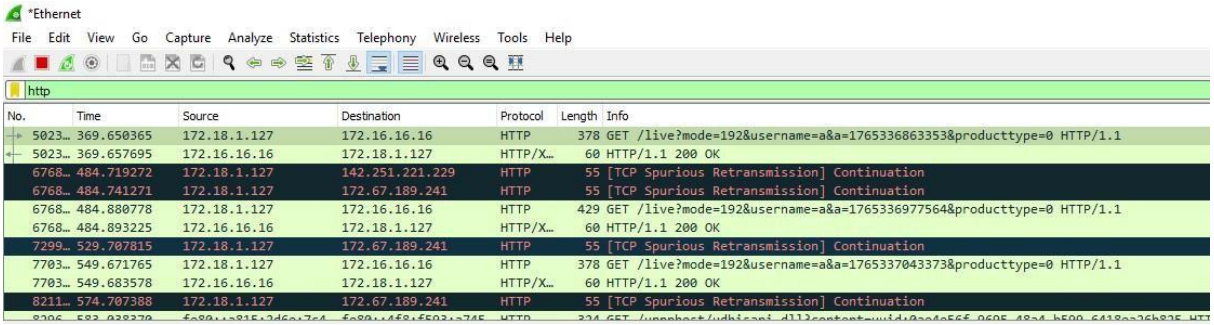


5. Now, click on get request and you can see the details.

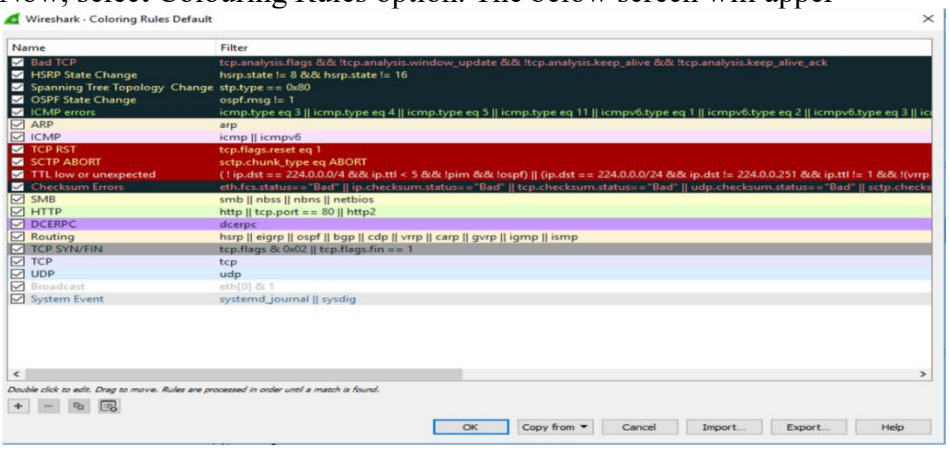


Colour Coding

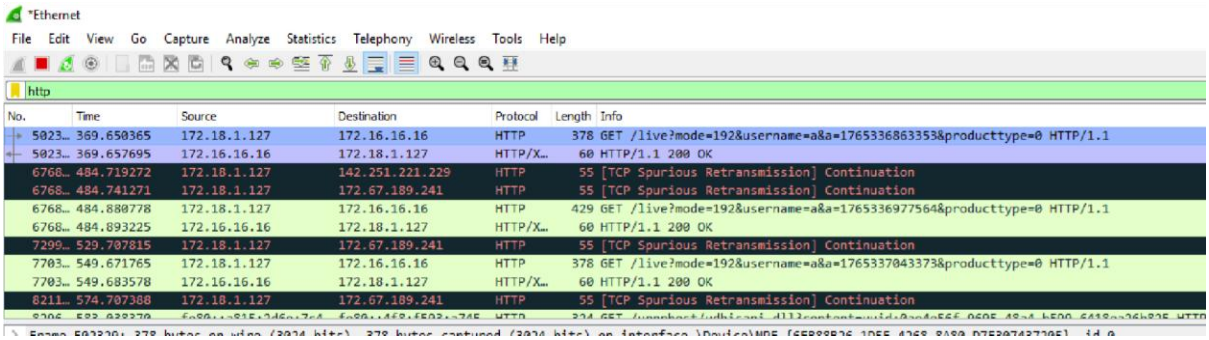
To check the colour coding rules click on View and select Colorize Packet List option. You will be able to see the below displayed output.



Now, select Colouring Rules option. The below screen will apper



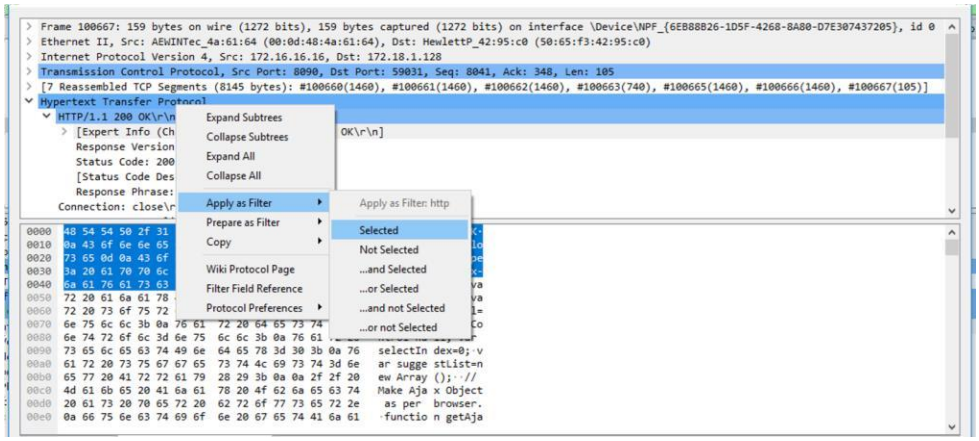
Select Colorize Conversation option -> Select any colour as per your wish. You will be able to see the below output.



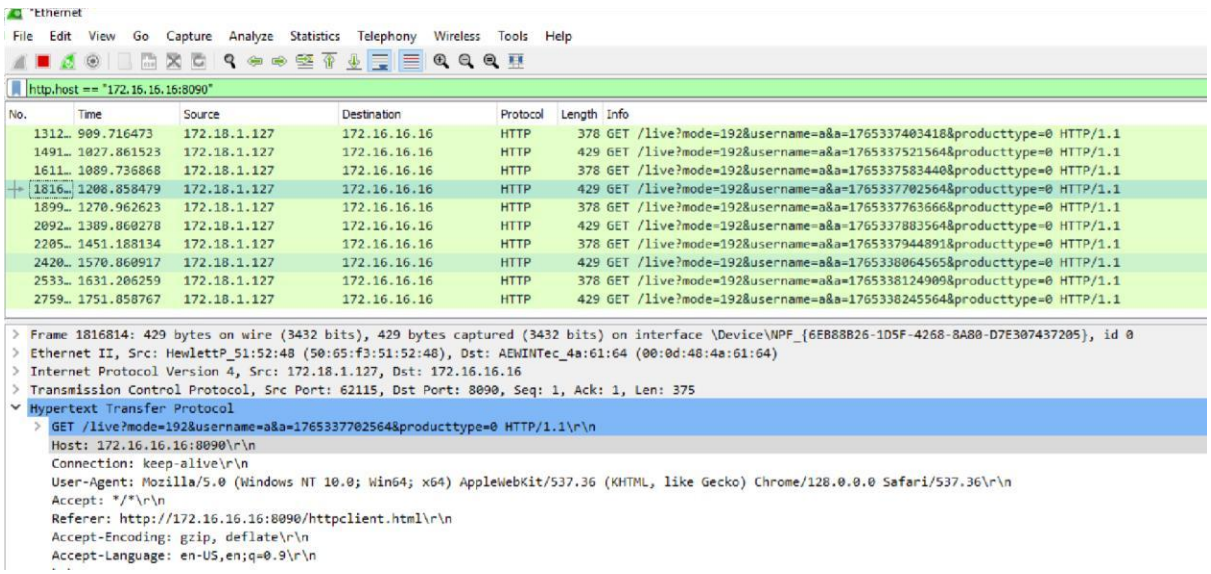
Analyse the captured Packets:

Initially, click on a packet and select it. Scroll down to view all its details.

Right click on any packet and select Apply filter -> Selected option.



After filtering, the following output will be displayed.



Statistics

Now, go to Statistics option -> Capture File Properties.

Details

File

Name:	C:\Users\Student\AppData\Local\Temp\wireshark_EthernetJO51G3.pcapng
Length:	6800MB
Hash (SHA256):	7f9b5bf1439cd5dc0e1ee8a1c852454536e9c27fff3d397f2c76b5c1f46b8cdc
Hash (RIPEMD160):	1552e2ea61a769d9ed4bf9dad7105184d652e717
Hash (SHA1):	34e2bace0515ec0cc8ca1ba6813042e93c123fc2
Format:	Wireshark/... - pcapng
Encapsulation:	Ethernet

Time

First packet:	2025-12-10 08:44:53
Last packet:	2025-12-10 09:37:42
Elapsed:	00:52:48

Capture

Hardware:	Intel(R) Core(TM) i3-4160 CPU @ 3.60GHz (with SSE4.2)
OS:	64-bit Windows 10 (1709), build 16299
Application:	Dumpcap (Wireshark) 3.4.8 (v3.4.8-0-g3e1ffae201b8)

Interfaces

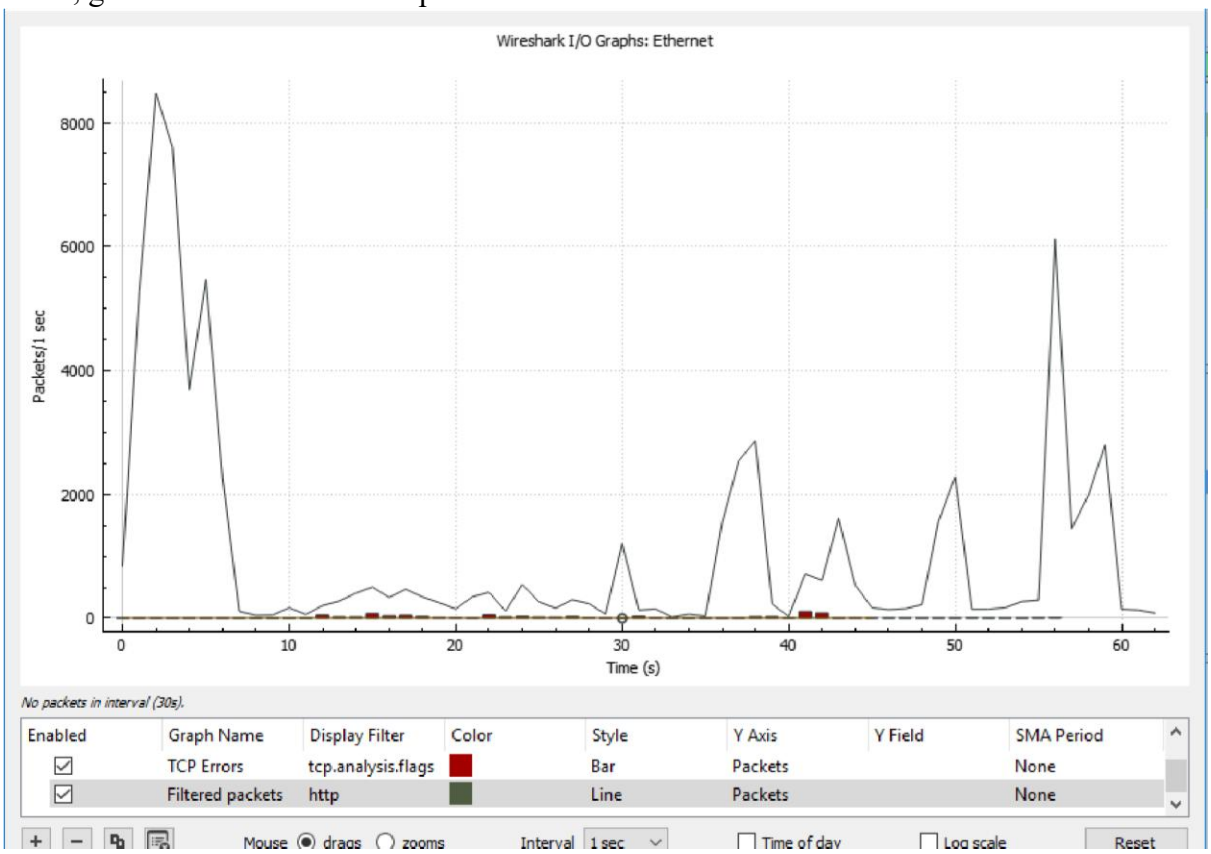
Interface	Dropped packets	Capture filter	Link type	Packet size limit
Ethernet	Unknown	none	Ethernet	262144 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	5401417	8569 (0.2%)	—
Time span, s	3168.334	3167.541	—
Average pps	1704.8	2.7	—
Average packet size, B	1225	138	—
Bytes	6616830517	1178374 (0.0%)	0
Average bytes/s	2088k	372	—

Capture file comments

Then, go to Statistics -> I/O Graphs.



Next, go to Statistics -> IPv4 Statistics -> All Addresses.

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
All Addresses	35337				0.4351	100%	5.3300	0.960
ff02::1	1				0.0000	0.00%	0.0100	31.323
8.8.8.8	365				0.0045	1.03%	0.2300	14.719
74.125.130.132	90				0.0011	0.25%	0.6200	18.804
72.154.7.248	14				0.0002	0.04%	0.0800	0.087
4.213.25.241	3				0.0000	0.01%	0.0300	13.104
255.255.255.255	113				0.0014	0.32%	0.0200	80.213
239.255.255.250	518				0.0064	1.47%	0.0500	79.981
239.192.152.143	3				0.0000	0.01%	0.0100	54.987
23.53.243.15	6				0.0001	0.02%	0.0400	10.600
225.16.8.68	1				0.0000	0.00%	0.0100	74.143
224.0.0.252	327				0.0040	0.93%	0.0800	15.208
224.0.0.251	22910				0.2821	64.83%	5.7800	3.775
224.0.0.22	173				0.0021	0.49%	0.1000	0.674
2001:0:348b:fb58:dc:25df:9879:63f2	1				0.0000	0.00%	0.0100	31.323
192.178.193.94	69				0.0008	0.20%	0.2900	78.625
172.18.3.28	8				0.0001	0.02%	0.0100	5.267
172.18.3.27	16				0.0002	0.05%	0.0100	3.828
172.18.3.255	282				0.0035	0.80%	0.0400	1.089
172.18.3.23	8				0.0001	0.02%	0.0100	9.159
172.18.3.203	1				0.0000	0.00%	0.0100	8.713

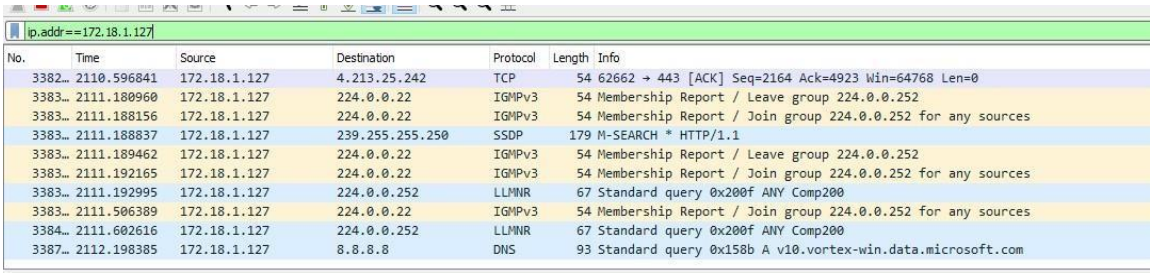
Finally, go to Statistics -> IPv6 Statistics -> All Addresses.

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
All Addresses	10592				0.1040	100%	1.3700	3.656
ff02::fb	8675				0.0852	81.90%	1.2600	3.656
ff02::c	207				0.0020	1.95%	0.0900	73.562
ff02::2	18				0.0002	0.17%	0.0200	7.338
ff02::1:ffd2:cead	1				0.0000	0.01%	0.0100	74.252
ff02::1:ffc6:333c	3				0.0000	0.03%	0.0100	7.398
ff02::1:ffb6:e423	3				0.0000	0.03%	0.0100	77.081
ff02::1:ff78:f8b	2				0.0000	0.02%	0.0100	73.907
ff02::1:ff6f:6a22	1				0.0000	0.01%	0.0100	0.314
ff02::1:ff6a:421f	7				0.0001	0.07%	0.0300	59.969
ff02::1:ff49:50e9	1				0.0000	0.01%	0.0100	72.274
ff02::1:ff21:b20d	1				0.0000	0.01%	0.0100	3.722
ff02::1:ff1e:9aaa	2				0.0000	0.02%	0.0100	34.555
ff02::1:ff1b:5ba2	1				0.0000	0.01%	0.0100	72.115
ff02::1:ff0d:d66a	1				0.0000	0.01%	0.0100	60.829
ff02::1:3	379				0.0037	3.58%	0.0600	3.737
ff02::1:2	22				0.0002	0.21%	0.0100	0.813
ff02::16	198				0.0019	1.87%	0.1000	0.673
ff02::1	4				0.0000	0.04%	0.0100	8.398
fe80::fe83:11c8:7d76:6875	404				0.0040	3.81%	0.0500	1.006
fe80::fd7b:9d07:b43c:aef4	8				0.0001	0.08%	0.0700	25.616

Display filter command

1. Display packets based on specific IP address

   ip.addr == 172.18.1.127

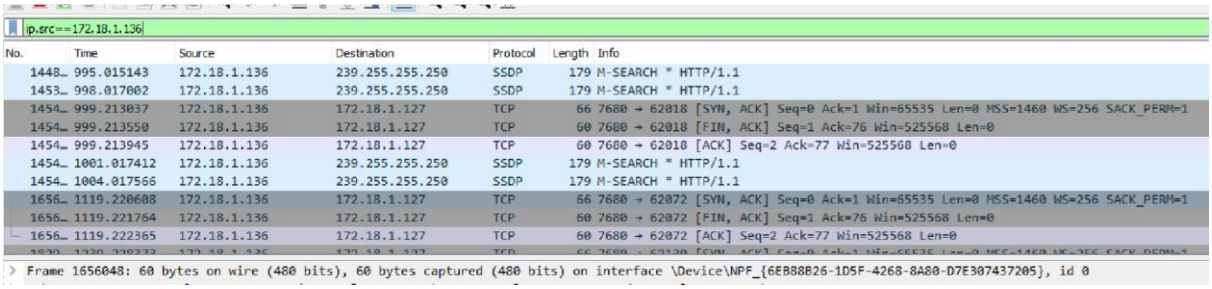


ip.addr==172.18.1.127

No.	Time	Source	Destination	Protocol	Length	Info
3382…	2110.596841	172.18.1.127	4.213.25.242	TCP	54	62662 → 443 [ACK] Seq=2164 Ack=4923 Win=64768 Len=0
3383…	2111.180960	172.18.1.127	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
3383…	2111.188156	172.18.1.127	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
3383…	2111.188837	172.18.1.127	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
3383…	2111.189462	172.18.1.127	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
3383…	2111.192165	172.18.1.127	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
3383…	2111.192995	172.18.1.127	224.0.0.252	LLMNR	67	Standard query 0x200f ANY Comp200
3383…	2111.506389	172.18.1.127	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
3384…	2111.602616	172.18.1.127	224.0.0.252	LLMNR	67	Standard query 0x200f ANY Comp200
3387…	2112.198385	172.18.1.127	8.8.8.8	DNS	93	Standard query 0x158b A v10.vortex-win.data.microsoft.com

2. Display packets which are coming from a specific IP address.

   ip.src == 172.18.1.136

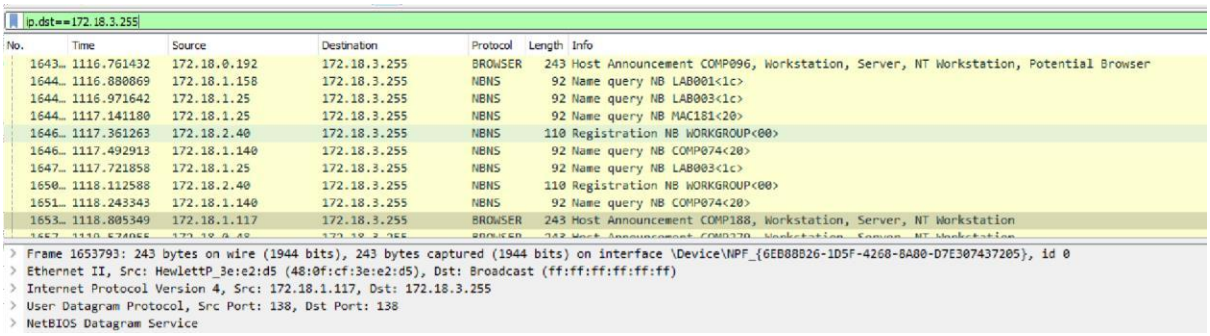


ip.src==172.18.1.136

No.	Time	Source	Destination	Protocol	Length	Info
1448…	995.015143	172.18.1.136	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1453…	998.017002	172.18.1.136	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1454…	999.213037	172.18.1.136	172.18.1.127	TCP	66	7680 → 62018 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1454…	999.213550	172.18.1.136	172.18.1.127	TCP	60	7680 → 62018 [FIN, ACK] Seq=1 Ack=76 Win=525568 Len=0
1454…	999.213945	172.18.1.136	172.18.1.127	TCP	60	7680 → 62018 [ACK] Seq=2 Ack=77 Win=525568 Len=0
1454…	1001.017412	172.18.1.136	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1454…	1004.017566	172.18.1.136	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1656…	1119.220608	172.18.1.136	172.18.1.127	TCP	66	7680 → 62072 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1656…	1119.221764	172.18.1.136	172.18.1.127	TCP	60	7680 → 62072 [FIN, ACK] Seq=1 Ack=76 Win=525568 Len=0
1656…	1119.222365	172.18.1.136	172.18.1.127	TCP	60	7680 → 62072 [ACK] Seq=2 Ack=77 Win=525568 Len=0

Frame 1656048: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{6EB88B26-1D5F-4268-8A80-D7E307437205}, id 0

3. Display packets which are having specific IP address destination.
   \ ip.dst == 172.18.3.255



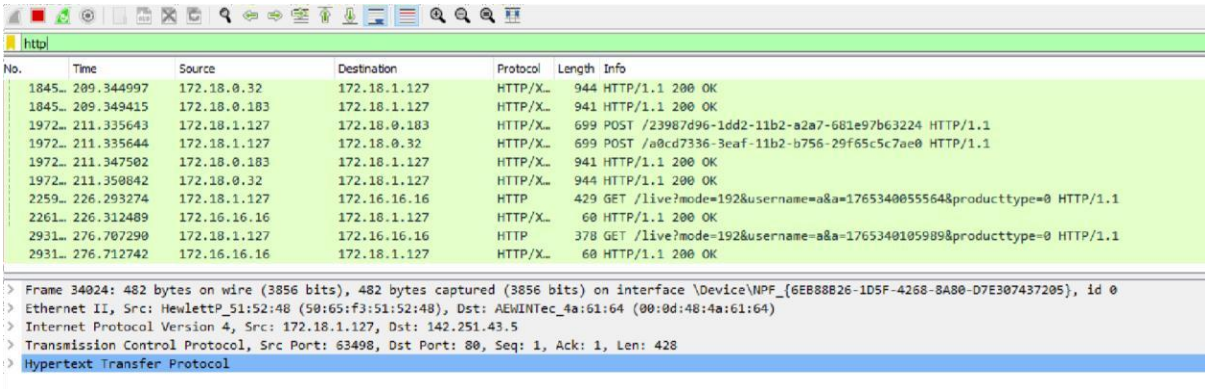
ip.dst==172.18.3.255

No.	Time	Source	Destination	Protocol	Length	Info
1643…	1116.761432	172.18.0.192	172.18.3.255	BROWSER	243	Host Announcement COMP096, Workstation, Server, NT Workstation, Potential Browser
1644…	1116.880869	172.18.1.158	172.18.3.255	NBNS	92	Name query NB LAB001<1c>
1644…	1116.971642	172.18.1.25	172.18.3.255	NBNS	92	Name query NB LAB003<1c>
1644…	1117.141180	172.18.1.25	172.18.3.255	NBNS	92	Name query NB MAC181<20>
1646…	1117.361263	172.18.2.40	172.18.3.255	NBNS	110	Registration NB WORKGROUP<00>
1646…	1117.492913	172.18.1.140	172.18.3.255	NBNS	92	Name query NB COMP074<20>
1647…	1117.721858	172.18.1.25	172.18.3.255	NBNS	92	Name query NB LAB003<1c>
1650…	1118.112588	172.18.2.40	172.18.3.255	NBNS	110	Registration NB WORKGROUP<00>
1651…	1118.243343	172.18.1.140	172.18.3.255	NBNS	92	Name query NB COMP074<20>
1653…	1118.805349	172.18.1.117	172.18.3.255	BROWSER	243	Host Announcement COMP188, Workstation, Server, NT Workstation

Frame 1653793: 243 bytes on wire (1944 bits), 243 bytes captured (1944 bits) on interface \Device\NPF_{6EB88B26-1D5F-4268-8A80-D7E307437205}, id 0
Ethernet II, Src: HewlettP_3e:e2:d5 (48:0f:cf:3e:e2:d5), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 172.18.1.117, Dst: 172.18.3.255
User Datagram Protocol, Src Port: 138, Dst Port: 138
NetBIOS Datagram Service

4. Display packets which are using http protocol.

   http



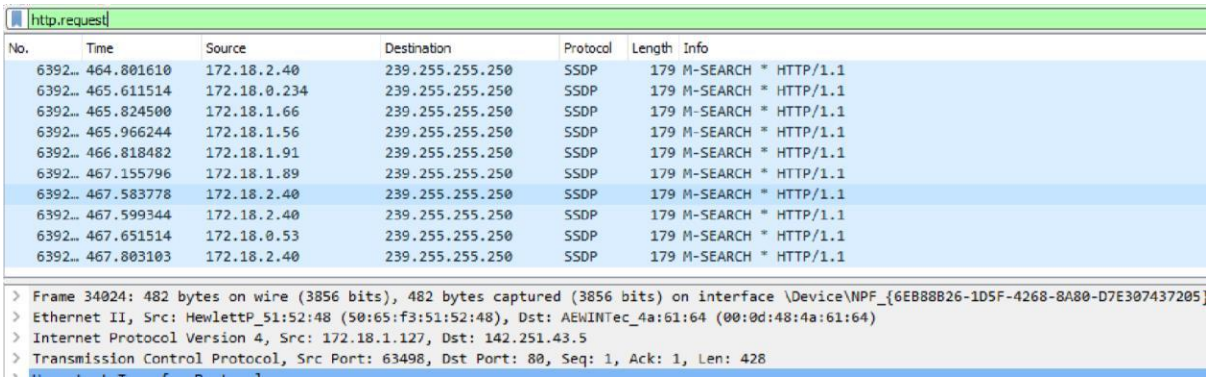
http

No.	Time	Source	Destination	Protocol	Length	Info
1845…	209.344997	172.18.0.32	172.18.1.127	HTTP/X…	944	HTTP/1.1 200 OK
1845…	209.349415	172.18.0.183	172.18.1.127	HTTP/X…	941	HTTP/1.1 200 OK
1972…	211.335643	172.18.1.127	172.18.0.183	HTTP/X…	699	POST /23987d96-1dd2-11b2-a2a7-681e97b63224 HTTP/1.1
1972…	211.335644	172.18.1.127	172.18.0.32	HTTP/X…	699	POST /a0cd7336-3eaf-11b2-b756-29f65c5c7ae0 HTTP/1.1
1972…	211.347502	172.18.0.183	172.18.1.127	HTTP/X…	941	HTTP/1.1 200 OK
1972…	211.350842	172.18.0.32	172.18.1.127	HTTP/X…	944	HTTP/1.1 200 OK
2259…	226.293274	172.18.1.127	172.16.16.16	HTTP	429	GET /live?mode=192&username=a&a=1765340055564&producttype=0 HTTP/1.1
2261…	226.312489	172.16.16.16	172.18.1.127	HTTP/X…	60	HTTP/1.1 200 OK
2931…	276.707290	172.18.1.127	172.16.16.16	HTTP	378	GET /live?mode=192&username=a&a=1765340105989&producttype=0 HTTP/1.1
2931…	276.712742	172.16.16.16	172.18.1.127	HTTP/X…	60	HTTP/1.1 200 OK

```
> Frame 34024: 482 bytes on wire (3856 bits), 482 bytes captured (3856 bits) on interface \Device\NPF_{6EB88B26-1D5F-4268-8A80-D7E307437205}, id 0
> Ethernet II, Src: HewlettP_51:52:48 (50:65:f3:51:52:48), Dst: AEWINTec_4a:61:64 (00:0d:48:4a:61:64)
> Internet Protocol Version 4, Src: 172.18.1.127, Dst: 142.251.43.5
> Transmission Control Protocol, Src Port: 63498, Dst Port: 80, Seq: 1, Ack: 1, Len: 428
> Hypertext Transfer Protocol
```

5. Display packets which are using http request.

   http.request



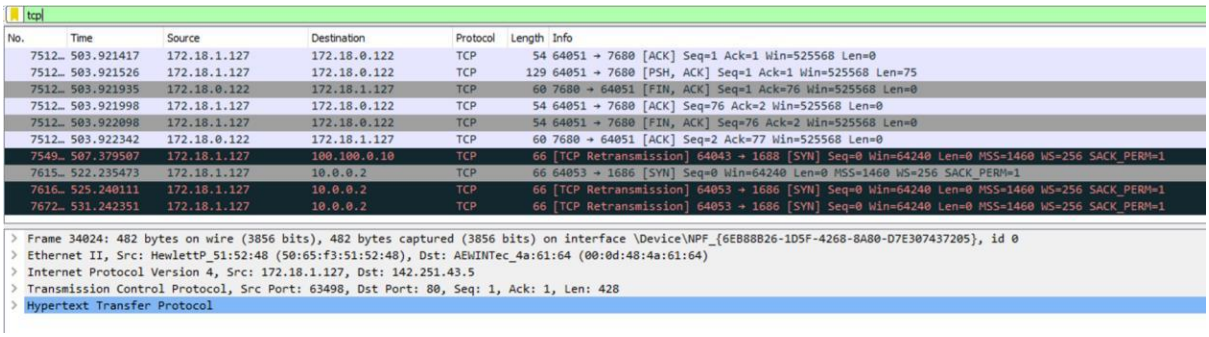
http.request

No.	Time	Source	Destination	Protocol	Length	Info
6392…	464.801610	172.18.2.40	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	465.611514	172.18.0.234	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	465.824500	172.18.1.66	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	465.966244	172.18.1.56	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	466.818482	172.18.1.91	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	467.155796	172.18.1.89	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	467.583778	172.18.2.40	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	467.599344	172.18.2.40	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	467.651514	172.18.0.53	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6392…	467.803103	172.18.2.40	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1

```
> Frame 34024: 482 bytes on wire (3856 bits), 482 bytes captured (3856 bits) on interface \Device\NPF_{6EB88B26-1D5F-4268-8A80-D7E307437205}
> Ethernet II, Src: HewlettP_51:52:48 (50:65:f3:51:52:48), Dst: AEWINTec_4a:61:64 (00:0d:48:4a:61:64)
> Internet Protocol Version 4, Src: 172.18.1.127, Dst: 142.251.43.5
> Transmission Control Protocol, Src Port: 63498, Dst Port: 80, Seq: 1, Ack: 1, Len: 428
```

6. Display packets which are using tcp protocol.

   tcp



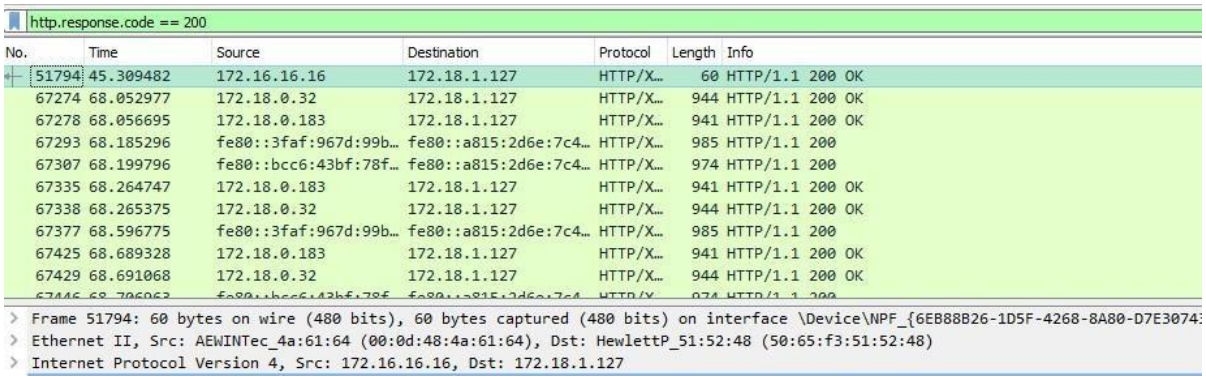
tcp

No.	Time	Source	Destination	Protocol	Length	Info
7512…	503.921417	172.18.1.127	172.18.0.122	TCP	54	64051 → 7680 [ACK] Seq=1 Ack=1 Win=525568 Len=0
7512…	503.921526	172.18.1.127	172.18.0.122	TCP	129	64051 → 7680 [PSH, ACK] Seq=1 Ack=1 Win=525568 Len=75
7512…	503.921935	172.18.0.122	172.18.1.127	TCP	60	7680 → 64051 [FIN, ACK] Seq=1 Ack=76 Win=525568 Len=0
7512…	503.921998	172.18.1.127	172.18.0.122	TCP	54	64051 → 7680 [ACK] Seq=76 Ack=2 Win=525568 Len=0
7512…	503.922098	172.18.1.127	172.18.0.122	TCP	54	64051 → 7680 [FIN, ACK] Seq=76 Ack=2 Win=525568 Len=0
7512…	503.922342	172.18.0.122	172.18.1.127	TCP	60	7680 → 64051 [ACK] Seq=2 Ack=77 Win=525568 Len=0
7549…	507.379507	172.18.1.127	100.100.0.10	TCP	66	[TCP Retransmission] 64043 → 1688 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7615…	522.235473	172.18.1.127	10.0.0.2	TCP	66	64053 → 1686 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7616…	525.240111	172.18.1.127	10.0.0.2	TCP	66	[TCP Retransmission] 64053 → 1686 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7672…	531.242351	172.18.1.127	10.0.0.2	TCP	66	[TCP Retransmission] 64053 → 1686 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1

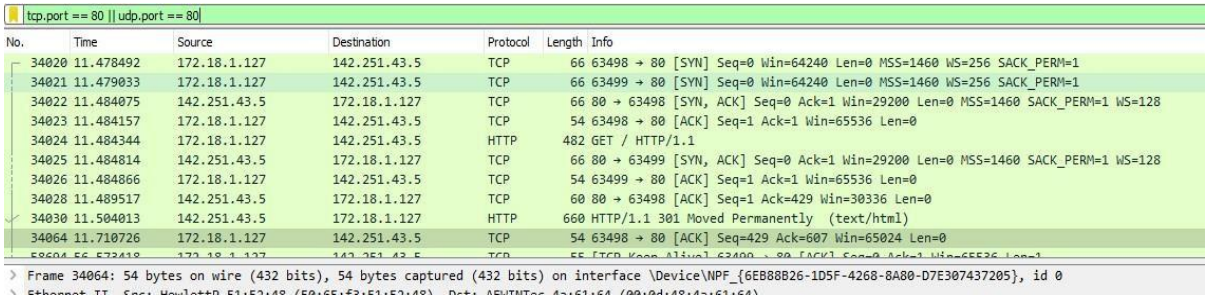
```
> Frame 34024: 482 bytes on wire (3856 bits), 482 bytes captured (3856 bits) on interface \Device\NPF_{6EB88B26-1D5F-4268-8A80-D7E307437205}, id 0
> Ethernet II, Src: HewlettP_51:52:48 (50:65:f3:51:52:48), Dst: AEWINTec_4a:61:64 (00:0d:48:4a:61:64)
> Internet Protocol Version 4, Src: 172.18.1.127, Dst: 142.251.43.5
> Transmission Control Protocol, Src Port: 63498, Dst Port: 80, Seq: 1, Ack: 1, Len: 428
> Hypertext Transfer Protocol
```

7. Display packets which are having no error with the

   connection. http.response.code == 200



8. Display packets which are having port

   number 80. tcp.port == 80 || udp.port

   == 80